

DTU





Model-based Machine Learning

Variational inference

Introduction

Variational inference

Variational Bayesian Expectation-Maximization

Road to MBML: where are we?

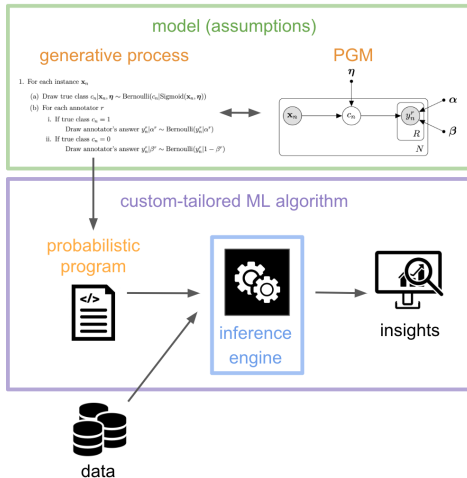
Foundations of PGMs and probabilistic programming

Probabilistic modelling in various contexts - your modelling toolbox

Bayesian inference (exact and approximate)

Week	Topic
1	Intro to the course + Prob. review
2	PGM foundations
3	PGM foundations II
4	Freq. vs Bayesian + Prob. Prog. + Mixture models
5	Regression models
6	Classification and Hierarchical models
7	Temporal models
8	Topic Models
9	Markov-chain Monte Carlo (MCMC)
10	Variational inference
11	Generative models
12	Gaussian processes
13	Project support

Model-based Machine Learning

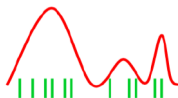


Previously, in MBML...

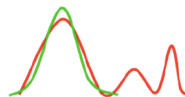
- Representation (lectures 2-4)
- Modelling toolbox (lectures 5-8, 11, 12)
- Inference (lectures 9 and 10)
 - Exact inference
 - Analytical (Bayes' rule)
 - Variable elimination
 - Belief propagation
 - Approximate inference
 - Stochastic methods - Markov chain Monte Carlo (MCMC)
 - Deterministic methods - Variational inference ^(\wedge)



True distribution



Monte Carlo



Variational

- Explain the concept of deterministic approximate inference.
- Explain variational inference.
- Describe differences between variational inference and expectation propagation.
- Derive and explain the evidence lower bound.
- Apply mean-field variational inference.
- Explain and apply Variational Bayesian Expectation-Maximization.

Approximate inference

- Suppose we wish to compute the posterior distribution of the latent variables $\mathbf{z}$ given some observed data $\mathbf{x}$: $p(\mathbf{z}|\mathbf{x})$
- For many problems of interest exact posterior inference is **intractable**

$$\begin{aligned}
 \overbrace{p(\mathbf{z}|\mathbf{x})}^{\text{posterior}} &= \frac{\overbrace{p(\mathbf{x}, \mathbf{z})}^{\text{joint}}}{\underbrace{p(\mathbf{x})}_{\text{evidence}}} = \frac{\overbrace{p(\mathbf{x}|\mathbf{z})}^{\text{likelihood}} \overbrace{p(\mathbf{z})}^{\text{prior}}}{\underbrace{p(\mathbf{x})}_{\text{evidence}}} = \frac{\overbrace{p(\mathbf{x}|\mathbf{z})}^{\text{likelihood}} \overbrace{p(\mathbf{z})}^{\text{prior}}}{\underbrace{\sum_{\mathbf{z}} p(\mathbf{x}|\mathbf{z}) p(\mathbf{z})}_{\text{evidence}}}
 \end{aligned}$$

- Cannot determine the posterior distribution analytically
- Cannot even compute expectations with respect to the posterior

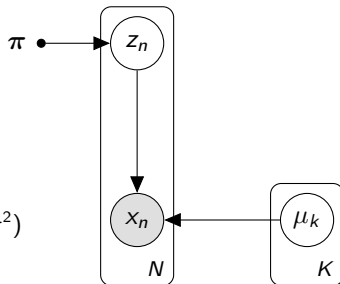
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 \end{aligned}$$

- Cannot determine the posterior distribution analytically
- Cannot even compute expectations with respect to the posterior
- Reasons for intractability:
 - Dimensionality of the latent space is too high to work with directly
 - Posterior distribution has a highly complex form for which expectations are not analytically tractable
- We must resort to **approximate inference** methods!

A practical example: Bayesian Gaussian mixture model



- For the sake of simplicity, assume π is given (fixed)
- ① For each cluster k :
 - a) Draw cluster center $\mu_k \sim \mathcal{N}(\mu_k|0, \tau^2)$
- ② For each data point $1, \dots, N$:
 - a) Draw cluster assignment $z_n \sim \text{Categorical}(z_n|\pi)$
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Note

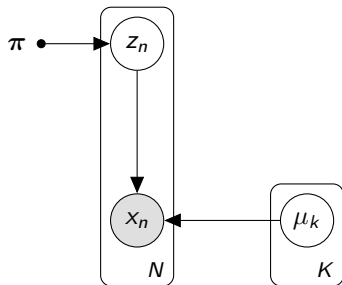
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A practical example: Bayesian Gaussian mixture model

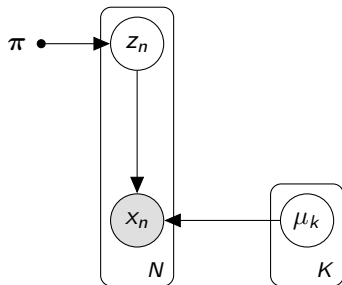
- Joint distribution is given by (ignoring the fixed parameters)

$$p(\boldsymbol{\mu}, \mathbf{z}, \mathbf{x}) = \left(\prod_{k=1}^K p(\mu_k) \right) \prod_{n=1}^N p(z_n | \pi) p(x_n | z_n, \boldsymbol{\mu})$$

where $\boldsymbol{\mu} = \{\mu_1, \dots, \mu_K\}$, $\mathbf{x} = \{x_1, \dots, x_N\}$ and $\mathbf{z} = \{z_1, \dots, z_N\}$



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- Our goal is to compute the posterior over $\boldsymbol{\mu}$ and $\mathbf{z}$

$$p(\boldsymbol{\mu}, \mathbf{z} | \mathbf{x}) = \frac{p(\boldsymbol{\mu}, \mathbf{z}, \mathbf{x})}{\int_{\boldsymbol{\mu}} \sum_{\mathbf{z}} p(\boldsymbol{\mu}, \mathbf{z}, \mathbf{x})}$$

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- The problem is the marginal likelihood in the denominator

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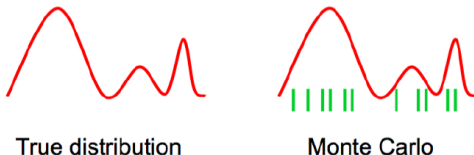
- Taking advantage of the conditional independence of the z_n 's:

$$p(\mathbf{x}) = \int_{\boldsymbol{\mu}} \left(\prod_{k=1}^K p(\mu_k) \right) \prod_{n=1}^N \sum_{z_n} p(z_n | \boldsymbol{\pi}) p(x_n | z_n, \boldsymbol{\mu})$$

- But this integral is still **intractable** to compute!

Approximate inference: stochastic methods

- Obtain a set of **samples** $\mathbf{z}^{(s)}$, for $s \in \{1, \dots, S\}$, drawn independently from the distribution $p(\mathbf{z}|\mathbf{x})$

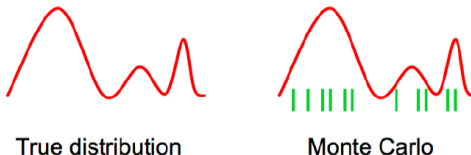


- Allows to approximate expectations as finite sums

$$\mathbb{E}_{p(\mathbf{z}|\mathbf{x})}[f(\mathbf{z})] \approx \frac{1}{S} \sum_{s=1}^S f(\mathbf{z}^{(s)})$$

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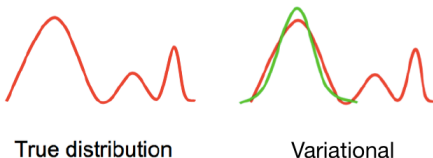
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- In the limit of infinite computational resources they can generate exact results
- Computationally demanding** and hard to scale to large datasets
- Many practical problems: determining convergence, number of samples, burn-in size, thinning, hard to diagnose, etc.

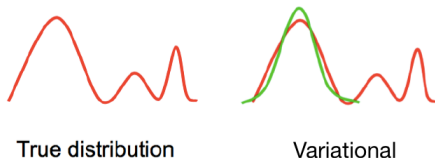
Approximate inference: variational methods

- Consider a family of tractable distributions $q(\mathbf{z}|\boldsymbol{\nu})$ - the **variational distribution**
- Find the **variational parameters** $\boldsymbol{\nu}$ that make $q(\mathbf{z}|\boldsymbol{\nu})$ as close as possible to the true posterior $p(\mathbf{z}|\mathbf{x})$



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- Turns the inference problem into an **optimization** problem!
- Use $q(\mathbf{z}|\boldsymbol{\nu})$ with the fitted parameters as a proxy for the true posterior
 - To make predictions about future data
 - To investigate the posterior distribution of the hidden variables

Mean-field variational inference

- How should we choose a **tractable** family of distributions for $q(\mathbf{z})$?

Mean-field variational inference

- How should we choose a **tractable** family of distributions for $q(\mathbf{z})$?
 - Relax some constraints in the true distribution (e.g. dependencies)
 - For example, we can assume a **fully factorized** approximation

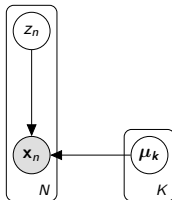
$$q(\mathbf{z}) = \prod_{m=1}^M q(z_m)$$

Mean-field variational inference

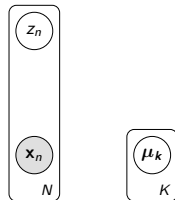
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$$q(\mathbf{z}) = \prod_{m=1}^M q(z_m)$$

- This is called a **mean-field approximation**



True joint distribution



Fully factorized approximation

- We can also assume that the distribution factorizes in groups of latent variables

Kullback-Leibler (KL) divergence

- What criteria defines the closeness between the two distributions?
 - **Kullback-Leibler (KL) divergence**

$$\mathbb{KL}(q(\mathbf{z})||p(\mathbf{z}|\mathbf{x})) = \int_{\mathbf{z}} q(\mathbf{z}) \log \frac{q(\mathbf{z})}{p(\mathbf{z}|\mathbf{x})} = \mathbb{E}_q \left[\log \frac{q(\mathbf{z})}{p(\mathbf{z}|\mathbf{x})} \right]$$

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- Notice that the KL divergence is an asymmetric measure

$$\mathbb{KL}(q(\mathbf{z})||p(\mathbf{z}|\mathbf{x})) \neq \mathbb{KL}(p(\mathbf{z}|\mathbf{x})||q(\mathbf{z}))$$

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- The reverse KL, $\mathbb{KL}(p(\mathbf{z}|\mathbf{x})||q(\mathbf{z}))$, gives rise to a different approximation inference algorithm called **expectation propagation (EP)**
 - It requires us to be able to take expectations with respect to $p(\mathbf{z}|\mathbf{x})$!
 - In general, it's more computationally expensive than variational inference
 - Watch http://videolectures.net/mlss09uk_minka_ai/?q=minka

Variational inference (VI) vs expectation propagation (EP)

- Different KL leads to different properties for the approximation
 - $\mathbb{KL}(p(\mathbf{z}|\mathbf{x})||q(\mathbf{z}))$ in EP is moment-matching
 - $\mathbb{KL}(q(\mathbf{z})||p(\mathbf{z}|\mathbf{x}))$ in VI is mode-seeking

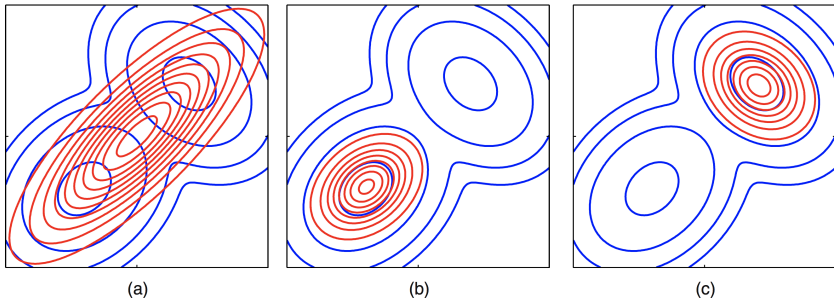


Figure: (a) $\mathbb{KL}(p||q)$. (b) $\mathbb{KL}(q||p)$. (c) Same as (b) but another local minimum. From Bishop (2006).

KL minimization

- Our goal is to find the variational parameters ν^* , such that

$$\nu^* = \arg \min_{\nu} \mathbb{KL}(q(\mathbf{z}|\nu) || p(\mathbf{z}|\mathbf{x})) = \arg \min_{\nu} \int_{\mathbf{z}} q(\mathbf{z}|\nu) \log \frac{q(\mathbf{z}|\nu)}{p(\mathbf{z}|\mathbf{x})}$$

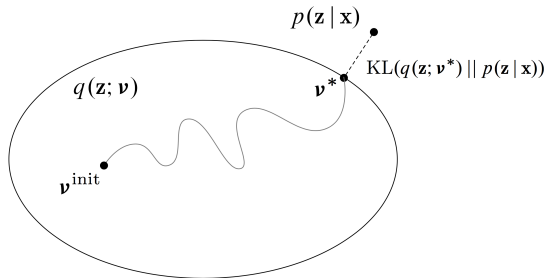


Figure: From David Blei (2017)

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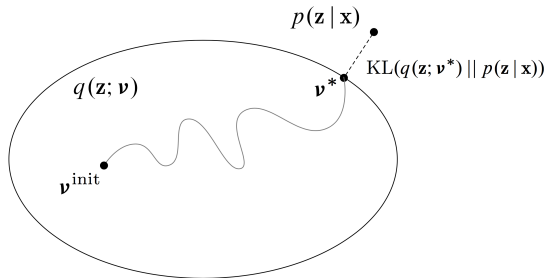


Figure: From David Blei (2017)

- Unfortunately, $\mathbb{KL}(q(\mathbf{z}) || p(\mathbf{z}|\mathbf{x}))$ cannot be minimized directly!

KL minimization

- However, we can find a function that we can minimize, which is equal to $\text{KL}(q(\mathbf{z})||p(\mathbf{z}|\mathbf{x}))$ up to an additive constant, as follows

$$\begin{aligned}\text{KL}(q(\mathbf{z})||p(\mathbf{z}|\mathbf{x})) &= \mathbb{E}_q \left[\log \frac{q(\mathbf{z})}{p(\mathbf{z}|\mathbf{x})} \right] \\ &= \mathbb{E}_q[\log q(\mathbf{z})] - \mathbb{E}_q[\log p(\mathbf{z}|\mathbf{x})] \\ &= \mathbb{E}_q[\log q(\mathbf{z})] - \mathbb{E}_q \left[\log \frac{p(\mathbf{z}, \mathbf{x})}{p(\mathbf{x})} \right] \\ &= - \underbrace{(\mathbb{E}_q[\log p(\mathbf{z}, \mathbf{x})] - \mathbb{E}_q[\log q(\mathbf{z})])}_{\mathcal{L}(q)} + \underbrace{\log p(\mathbf{x})}_{\text{const.}}\end{aligned}$$

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- The $\log p(\mathbf{x})$ term does not depend on q and thus it can be ignored
- Minimizing the KL divergence is then equivalent to maximizing $\mathcal{L}(q)$
- $\mathcal{L}(q)$ is called the **evidence lower bound (ELBO)**

KL minimization

- The ELBO, $\mathcal{L}(q)$, is a **lower bound** on the log model evidence $\log p(\mathbf{x})$

$$\text{KL}(q(\mathbf{z})||p(\mathbf{z}|\mathbf{x})) = \underbrace{-\left(\mathbb{E}_q[\log p(\mathbf{z}, \mathbf{x})] - \mathbb{E}_q[\log q(\mathbf{z})]\right)}_{\text{ELBO } \mathcal{L}(q)} + \underbrace{\log p(\mathbf{x})}_{\text{log evidence}}$$

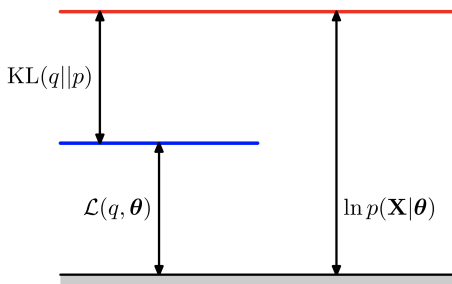


Figure: From Bishop (2006)

- The ELBO $\mathcal{L}(q)$ is tight when $q(\mathbf{z}) \approx p(\mathbf{z}|\mathbf{x})$, in which case $\mathcal{L}(q) \approx \log p(\mathbf{x})$

- The ELBO, $\mathcal{L}(q)$, is a **lower bound** on the log model evidence $\log p(\mathbf{x})$

$$\begin{aligned}\log p(\mathbf{x}) &= \log \int_{\mathbf{z}} p(\mathbf{z}, \mathbf{x}) \\ &= \log \int_{\mathbf{z}} q(\mathbf{z}) \frac{p(\mathbf{z}, \mathbf{x})}{q(\mathbf{z})} \\ &= \log \mathbb{E}_q \left[\frac{p(\mathbf{z}, \mathbf{x})}{q(\mathbf{z})} \right] \\ &\geq \underbrace{\mathbb{E}_q[\log p(\mathbf{z}, \mathbf{x})] - \mathbb{E}_q[\log q(\mathbf{z})]}_{\mathcal{L}(q)}\end{aligned}$$

- The last step comes from **Jensen's inequality**

$$\log \mathbb{E}[p(\mathbf{x})] \geq \mathbb{E}[\log p(\mathbf{x})]$$

- This is a consequence of the concavity of the logarithmic function

- Our goal is to find the variational parameters ν that maximize the ELBO $\mathcal{L}(q)$

$$\mathcal{L}(q) = \mathbb{E}_q[\log p(\mathbf{z}, \mathbf{x})] - \mathbb{E}_q[\log q(\mathbf{z}|\nu)]$$

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- Can be maximized using a **coordinate ascent algorithm**
 - Iteratively optimize the variational parameters of each latent variable $q(z_m)$ in turn, holding the others fixed, until a convergence criteria is met
- Ensures convergence to a local maximum of $\mathcal{L}(q)$
- Remember, at convergence:

$$q(\mathbf{z}) \approx p(\mathbf{z}|\mathbf{x})$$

$$\mathcal{L}(q) \approx \log p(\mathbf{x})$$

Note

Variational inference (VI) is also often called **Variational Bayes (VB)**.

Two different perspectives on the ELBO $\mathcal{L}(q)$

- Expected value of log joint probability minus entropy terms

$$\mathcal{L}(q) = \mathbb{E}_q[\log p(\mathbf{z}, \mathbf{x})] - \underbrace{\mathbb{E}_q[\log q(\mathbf{z}|\boldsymbol{\nu})]}_{\text{entropy terms}}$$

- Entropy terms favour less informative variational distributions $q(\mathbf{z}|\boldsymbol{\nu})$

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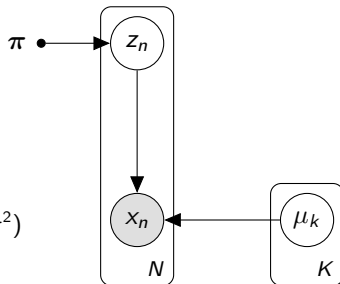
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- Entropy terms favour less informative variational distributions $q(\mathbf{z}|\nu)$
- Expected value of log likelihood minus KL divergence to prior

$$\begin{aligned}\mathcal{L}(q) &= \mathbb{E}_q[\log p(\mathbf{x}|\mathbf{z})] - \mathbb{E}_q[\log p(\mathbf{z})] - \mathbb{E}_q[\log q(\mathbf{z}|\nu)] \\ &= \mathbb{E}_q[\log p(\mathbf{x}|\mathbf{z})] - \underbrace{\mathbb{E}_q\left[\log \frac{q(\mathbf{z}|\nu)}{p(\mathbf{z})}\right]}_{\text{KL}(q(\mathbf{z}|\nu)||p(\mathbf{z}))}\end{aligned}$$

- $\mathbb{E}_q[\log p(\mathbf{x}|\mathbf{z})]$ is the reconstruction loss
- $\text{KL}(q(\mathbf{z}|\nu)||p(\mathbf{z}))$ acts like a regularization term (to stay close to the prior)

A practical example: Bayesian Gaussian mixture model

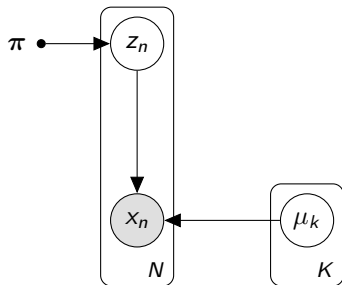


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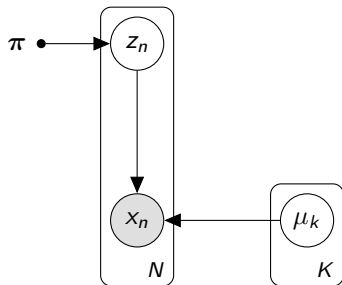
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- Approximate (mean-field) distribution:

$$q(\boldsymbol{\mu}, \mathbf{z}) =$$

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- Approximate (mean-field) distribution:

$$q(\boldsymbol{\mu}, \mathbf{z}) = \left(\prod_{k=1}^K q(\mu_k | \tilde{\mu}_k, \tilde{\sigma}_k) \right) \prod_{n=1}^N q(z_n | \phi_n)$$

A practical example: Bayesian Gaussian mixture model

- Our goal is to find the variational parameters that maximize the ELBO $\mathcal{L}(q)$

$$\mathcal{L}(q) = \mathbb{E}_q[\log p(\boldsymbol{\mu}, \mathbf{z}, \mathbf{x})] - \mathbb{E}_q[\log q(\boldsymbol{\mu}, \mathbf{z})]$$

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$$\mathcal{L}(q) = \mathbb{E}_q[\log p(\boldsymbol{\mu}, \mathbf{z}, \mathbf{x})] - \mathbb{E}_q[\log q(\boldsymbol{\mu}, \mathbf{z})]$$

- Because q is fully factorized, the expectations in $\mathcal{L}(q)$ decompose into sums of simpler terms

$$\begin{aligned}\mathbb{E}_q[\log p(\boldsymbol{\mu}, \mathbf{z}, \mathbf{x})] &= \sum_{k=1}^K \mathbb{E}_q[\log p(\mu_k | 0, \tau^2)] + \sum_{n=1}^N \mathbb{E}_q[\log p(z_n | \boldsymbol{\pi})] \\ &\quad + \sum_{n=1}^N \mathbb{E}_q[\log p(x_n | z_n, \boldsymbol{\mu}, \sigma^2)]\end{aligned}$$

$$\mathbb{E}_q[\log q(\boldsymbol{\mu}, \mathbf{z})] = \sum_{k=1}^K \mathbb{E}_q[\log q(\mu_k | \tilde{\mu}_k, \tilde{\sigma}_k)] + \sum_{n=1}^N \mathbb{E}_q[\log q(z_n | \phi_n)]$$

Expectations, expectations, expectations...

- For a discrete variable X :

$$\mathbb{E}[f(X)] = \sum_X p(X) f(X)$$

- For a continuous variable X :

$$\mathbb{E}[f(X)] = \int p(X) f(X) dX$$

- Some useful properties of expectations:

$$\mathbb{E}[a] = a$$

$$\mathbb{E}[a + bX] = a + b \mathbb{E}[X]$$

$$\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$$

$$\mathbb{E}[XY] = \mathbb{E}[X] \mathbb{E}[Y], \quad \text{only if } X \text{ and } Y \text{ are independent}$$

$$\mathbb{V}[X] = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - \mathbb{E}[X]^2$$

- More details on expectations:

<https://www3.nd.edu/~rwilliam/stats1/x12.pdf>

Expectations, expectations, expectations...

- Consider the approximate (mean-field) distribution:

$$q(\boldsymbol{\mu}, \mathbf{z}) = \left(\prod_{k=1}^K \mathcal{N}(\mu_k | \tilde{\mu}_k, \tilde{\sigma}_k) \right) \prod_{n=1}^N \text{Categorical}(z_n | \phi_n)$$

- Using the properties of expectations and the factorization of $q(\boldsymbol{\mu}, \mathbf{z})$:

$$\mathbb{E}_q[\mu_k] =$$

$$\mathbb{E}_q[\mu_k^2] =$$

$$\mathbb{E}_q[z_n] =$$

$$\mathbb{E}_q[z_{n,k}] =$$

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Expectations, expectations, expectations...

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Playtime!

- **Calculate and implement** the coordinate ascent updates for maximizing $\mathcal{L}(q)$

$$\mathcal{L}(q) = \mathbb{E}_q[\log p(\boldsymbol{\mu}, \mathbf{z}, \mathbf{x})] - \mathbb{E}_q[\log q(\boldsymbol{\mu}, \mathbf{z})]$$

- Jupyter notebook: "10 - Variational inference - Bayesian GMM.ipynb"
- Hints:
 - Consider each variational parameter in turn: $\tilde{\mu}_k$, $\tilde{\sigma}_k$ and $\phi_{n,k}$

Table: Variables in the model and corresponding variational parameters

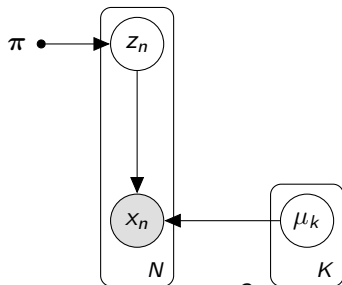
Model variable	μ_k		z_n
Variational parameter	$\tilde{\mu}_k$	$\tilde{\sigma}_k$	ϕ_n

- For each variational parameter, keep the relevant terms in $\mathcal{L}(q)$
- Take the derivative with respect to the variational parameter and set it to zero
- If necessary, use Lagrange multipliers to ensure constraints (e.g. $\sum_{k=1}^K \phi_{n,k} = 1$)

Note

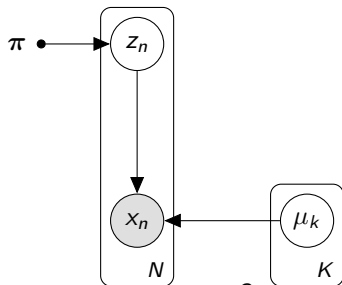
Black-box variational inference in Pyro attempts to do all of this automatically (sometimes successfully, other times not so much)

Optimizing hyper-parameters



- What if we wanted to optimize the hyper-parameters π too?

Optimizing hyper-parameters



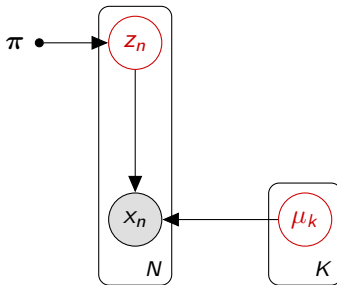
- What if we wanted to optimize the hyper-parameters π too?
- ELBO $\mathcal{L}(q)$ is a lower bound on the log marginal likelihood of the data $\log p(\mathbf{x})$

$$\log p(\mathbf{x}) = \log \int_{\mu} \left(\prod_{k=1}^K p(\mu_k) \right) \prod_{n=1}^N \sum_{z_n} p(z_n | \pi) p(x_n | z_n, \mu) \geq \mathcal{L}(q)$$

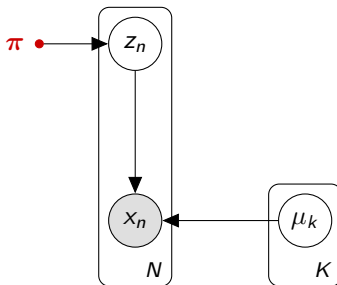
- At convergence of VI, this bound is tight: $\mathcal{L}(q) \approx \log p(\mathbf{x})$
- We can use $\mathcal{L}(q)$ to find a maximum likelihood estimate of π !

- Goal is to find posterior over latent variables $\mathbf{z}$ using VI and maximum likelihood estimates for hyper-parameters θ

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 - E-step: run VI to find approximate posterior $q(\mathbf{z})$, such that $q(\mathbf{z}) \approx p(\mathbf{z}|\mathbf{x})$
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Relation with standard Expectation-Maximization (EM)

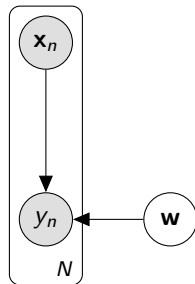
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Note

If you are familiar with the popular Expectation-Maximization (EM) algorithm, it is a special case of VB-EM, when the E-step is exact (exact inference instead of VI).

VI example: Automatic Relevance Determination (ARD)

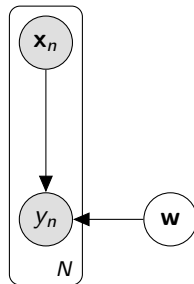
- Recall the Bayesian linear regression model:
 - 1 Draw weights $\mathbf{w} \sim \mathcal{N}(\mathbf{w}|\mathbf{0}, \alpha^{-1}\mathbf{I})$
 - 2 For each data point $n \in \{1, \dots, N\}$
 - a Draw target $y_n \sim \mathcal{N}(y_n|\mathbf{w}^T \mathbf{x}_n, \sigma^2)$
- It assumes independent priors on the weights $\mathbf{w}$ with the same variance (α^{-1}) for all input dimensions $d \in \{1, \dots, D\}$



VI example: Automatic Relevance Determination (ARD)

- Recall the Bayesian linear regression model:

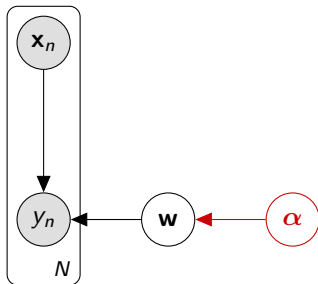
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- It assumes independent priors on the weights $\mathbf{w}$ with the same variance (α^{-1}) for all input dimensions $d \in \{1, \dots, D\}$
- Alternatively, we consider a different variance α_d^{-1} for each input dimension
- We can then specify a prior on the precisions $\alpha_d \sim \text{Gamma}(\alpha_d|a_0, b_0)$
- This is called an **hyper-prior**!

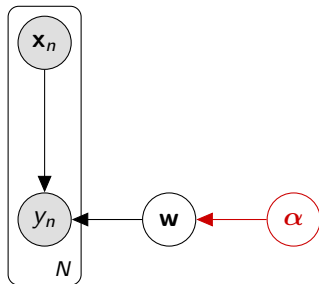
VI example: Automatic Relevance Determination (ARD)

- New generative process and PGM:
 - For each input dimension $d \in \{1, \dots, D\}$
 - Draw weight precision
 $\alpha_d \sim \text{Gamma}(\alpha_d | a_0, b_0)$
 - Draw weights $\mathbf{w} \sim \mathcal{N}(\mathbf{w} | \mathbf{0}, \alpha^{-1} \mathbf{I})$
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- Prior on α_d specifies how uncertain we are a-priori that these weights are small



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 - a Draw target $y_n \sim \mathcal{N}(y_n | \mathbf{w}^T \mathbf{x}_n, \sigma^2)$
- Prior on α_d specifies how uncertain we are a-priori that these weights are small
- Posterior on α_d tells us how relevant the d^{th} dimension is - hence the name ARD!
- Added flexibility allows some weights w_d to be further pushed towards zero
- It allows to do **automatic feature selection**!



VI example: Automatic Relevance Determination (ARD)

- The joint distribution is given by:

$$p(\mathbf{y}, \mathbf{w}, \boldsymbol{\alpha} | \mathbf{X}) = \mathcal{N}(\mathbf{w} | \mathbf{0}, \boldsymbol{\alpha}^{-1} \mathbf{I}) \left(\prod_{j=1}^D \text{Ga}(\alpha_j | a_0, b_0) \right) \prod_{n=1}^N \mathcal{N}(y_n | \mathbf{w}^T \mathbf{x}_n, \sigma^2)$$

where $\mathbf{y} = \{y_1, \dots, y_N\}$ and $\mathbf{X} = \{\mathbf{x}_1, \dots, \mathbf{x}_N\}$

VI example: Automatic Relevance Determination (ARD)

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- Exact inference is unfortunately now intractable

$$p(\mathbf{w}, \boldsymbol{\alpha} | \mathbf{y}, \mathbf{X}) = \frac{p(\mathbf{y}, \mathbf{w}, \boldsymbol{\alpha} | \mathbf{X})}{\int_{\mathbf{w}} \int_{\boldsymbol{\alpha}} p(\mathbf{y}, \mathbf{w}, \boldsymbol{\alpha} | \mathbf{X})}$$

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- But we can use (mean-field) variational inference!

$$q(\mathbf{w}, \boldsymbol{\alpha}) = \mathcal{N}(\mathbf{w} | \mathbf{m}_N, \mathbf{V}_N) \prod_{j=1}^D \text{Ga}(\alpha_j | a_{Nj}, b_{Nj})$$

- Our goal is to find the variational parameters $\mathbf{m}_N$, $\mathbf{V}_N$, a_{Nj} and b_{Nj} that maximize the ELBO $\mathcal{L}(q)$:

$$\mathcal{L}(q) = \mathbb{E}_q[\log p(\mathbf{y}, \mathbf{w}, \boldsymbol{\alpha} | \mathbf{X})] - \mathbb{E}_q[\log q(\mathbf{w}, \boldsymbol{\alpha})]$$

- **Implement** the coordinate ascent updates for maximizing $\mathcal{L}(q)$

$$\mathcal{L}(q) = \mathbb{E}_q[\log p(\mathbf{y}, \mathbf{w}, \alpha | \mathbf{X})] - \mathbb{E}_q[\log q(\mathbf{w}, \alpha)]$$

- Jupyter notebook: "10 - Variational inference - ARD.ipynb"
- Hints:
 - Consider each variational parameter in turn: $\mathbf{m}_N$, $\mathbf{V}_N$, a_{Nj} and b_{Nj}

Table: Variables in the model and corresponding variational parameters

Model variable	$\mathbf{w}$		α_j	
Variational parameter	$\mathbf{m}_N$	$\mathbf{V}_N$	a_{Nj}	b_{Nj}

Note

Notice how fast this VI algorithm is when compared to MCMC inference in STAN!